Darshan University

A Project Report on

"Hotel Management System"

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Software Engineering (2301CS405)

B. Tech, Semester – IV

Computer Science & Engineering Department

Submitted By

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Internal Guide

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DECLARATION

We hereby declare that the SRS, submitted along with the **Software Engineering (2301CS405)** for entitled **"Hotel Management System"** submitted in partial fulfilment for the Semester-5 of **Bachelor Technology (B. Tech)** in **Computer Science and Engineering (CSE)** Department to Darshan University, Rajkot, is a record of the work carried out at **Darshan University, Rajkot** under the supervision of R. B. Gondaliya and that no part of any of report has been directly copied from any students' reports, without providing due reference.

PUJAN

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CERTIFICATE

This is to certify that the SRS on **"Hotel Management System"** has been satisfactorily prepared by **Pujan (23010101426)** under my guidance in the fulfillment of the course **Software Engineering (2301CS405)** work during the academic year 2024-2025.

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Thus, in conclusion to the above said, I once again thank the faculties and members of **Darshan University** for their valuable support in completion of the project.

Thanking You

Pujan Pipaliya

ABSTRACT

A Hotel Management System (HMS) is a comprehensive software solution designed to streamline the operations of a hotel, improving the efficiency of both front-end and back-end processes. The system automates key hotel management tasks such as guest reservations, check-in/check-out procedures, billing, room management, and housekeeping coordination. It provides a user-friendly interface for hotel staff to manage room availability, process payments, track customer preferences, and generate reports for analysis.

The primary objectives of the Hotel Management System are to enhance customer service, reduce manual errors, and increase operational efficiency. By integrating various functions like reservation management, guest tracking, room scheduling, and inventory control, the system ensures smooth hotel operations and improved customer satisfaction.

With the incorporation of features such as online booking, real-time availability updates, and customizable billing options, the HMS caters to both hotel staff and guests. The system also includes modules for accounting, payroll, and reporting to support managerial decision-making.

In summary, a Hotel Management System offers an automated, integrated approach to managing hotel operations, ensuring better resource management, improved customer experience, and overall business growth.

Table of Contents

1 Student Name: Pipaliya Pujan 1
2 EnrollmentNo.:23010101426 1
3 List of Figures I
List of Tables II
4 Introduction 1
4.1 Product perspective 1
4.2 Product features 1
4.2.1 There are three different users who will be using this product: 1
4.2.2 The features that are required for the Manager are: 1
4.2.3 The features that are required for the Customer are: 1
4.3 Functional Requirement 1
4.3.1 Manager 1
4.3.2 Customer 2
4.3.3 Receptionist 3
4.3.4 Payment Handler 3
4.4 Non-Functional Requirement 4
5 Design and Implementation Constraints 5
5.1 Use case diagram 5
5.2 Activity diagram and Swimlane diagram 6
5.3 Sequence diagram 9
5.4 State diagram 11
5.5 Class diagram 12
5.6 Data flow diagram 13
5.6.1 Context diagram (level-0) 13
5.6.2 DFD Level-1 13
6 External interface requirement (Screens) 14
6.1 Screen-1: Create Employees 14
6.2 Screen-2: Set Password 15
6.3 Screen-3: Customer using Barcode Scanner 16
6.4 Screen-4: Payment Processing for Customer: 17
6.5 Screen-5: Cancel a Booking: 18
7 Database design 20
7.1 List of Tables 20
8 Stories and Scenario 22
8.1 Story-1: Book a Room in Hotel Catalogue 22
8.1.1 Scenario# S1.1 22
8.1.2 Scenario# S1.2 22

8.2 Story-2: Available rooms 23
8.2.1 Scenario# S2.1 23
8.2.2 Scenario# S2.2 23
8.3 Story-3: Make Payment 24
8.3.1 Scenario# S3.1 24
8.3.2 Scenario# S3.2 24
9 Test cases 25
10 References 29

3 List of Figures

Figure 5.1- 1 Use case diagram for Hotel management system 5
Figure 5.2- 1 Activity diagram for Room Booking 6
Figure 5.2- 2 Swimlane diagram for Room Booking 6
Figure 5.3- 1 Sequence diagram for Room Booking 9
Figure 5.3- 2 Sequence diagram for Payment 10
Figure 5.4- 1 State diagram of Room 11
Figure 5.4- 2 State diagram for Customer 11
Figure 5.5- 1 Class diagram for Hotel management system 12
Figure 5.6- 1 Context diagram for Hotel management system 13
Figure 5.6- 2 DFD level-1 for Hotel management system 13
Figure 6.1- 1 Screen-1: Create Employees 14
Figure 6.1- 2 Screen-2: Set password 15
Figure 6.1- 3 Screen-3: Add Customer 16
Figure 6.1-4 Screen-4: Payment Processing for Customer 16
Figure 6.1-5 Screen-5: Cancel a Booking 17

List of Tables

Table 6.1- 1 Screen element of Create Employees 14
Table 6.1- 2 Screen element of Set Password 15
Table 6.1- 3 Screen element of Add Customer 16
Table 6.1- 4 Screen element of Payment 17
Table 6.1- 5 Screen element Cancel Booking 18
Table 7.1- 1 Table: Room 20
Table 7.1- 2 Table: Booking 20
Table 7.1- 3 Table: Customer 20
Table 7.1- 4 Table: Payment 21

4 Introduction

4.1 Product perspective

The Hotel Management System is a centralized software that automates hotel operations like reservations, billing, and guest management. It offers real-time room availability and integrates with services like POS, accounting, and CRM. Designed for multiple user roles, it enhances operational efficiency and guest experience, making it essential for modern hotels.

4.2 Product features

4.2.1 There are three different users who will be using this product:

- Manager who will be acting as the administrator.
- Receptionist who will be accessing the Rooms View.
- Customer who will request for Room Booking.

4.2.2 The features that are required for the Manager are:

- **Analytics & Reporting**: View detailed reports on occupancy, revenue, guest demographics and performance metrics.
- **Staff Management**: Manage employee schedules, tasks, and payroll.
- **Financial Management**: Track overall hotel revenue, expenses, and budget forecasting.
- **Inventory Management**: Monitor stock levels for supplies, equipment, and F&B services.
- **Room Rate Management**: Set and update room rates based on demand, season, and promotions.
- **Guest Relationship Management**: Track guest preferences and ensure high-quality service for repeat visitors.

4.2.3 The features that are required for the Customer are:

- **Room Booking**: Customers can book rooms online or via the front desk, with real-time availability updates.
- **Reservation Modification**: Modify or cancel bookings easily through the hotel website or at the reception.
- **Check-in/Check-out**: Streamlined check-in and check-out process with digital keys (if available) for contactless service.

4.3 Functional Requirement

4.3.1 <u>Manager</u>

- **Edit Employee Information**: The manager can update employee details, such as contact info, position, and work hours.
- **Create Employees**: The manager can add new employees (staff, admin, etc.) to the system with relevant details (name, role, contact).
- **Assign Roles and Permissions**: The manager can define roles (e.g., admin, receptionist, housekeeper) and set permissions accordingly.
- **Set Passwords**: The manager can assign and reset passwords for employees to ensure secure access to the system.

- **Access Level Management**: The manager can control which employees have access to specific features (e.g., financial data, reports, room bookings).
- **Set Operational Rules**: The manager can define rules related to hotel operations, such as booking policies, check-in/check-out times, cancellation rules, and payment terms.
- **Manage Special Requests**: The manager should be able to track and fulfill special guest requests (e.g., early check-in, additional services).
- **Revenue Reports**: The manager can view reports related to room revenue, additional services, and total income.
- **Employee Scheduling**: The manager can create and manage staff schedules, including housekeeping, reception, and other hotel staff.
- **Set and Update Hotel Policies**: The hotel manager can update policies related to cancellations, no-shows, check-in/check-out times, and pet-friendly rules.
- **Room Access Control**: The manager should be able to control room access, such as deactivating or activating key cards.
- **Rate Management**: The hotel manager should be able to set and adjust room rates based on season, demand, or special promotions.

4.3.2 Customer

- **Document Upload**: Customers can upload necessary documents (ID, passport) via the system during check-in for verification.
- **Barcode/QR Code Scan**: Customers can check in by scanning a barcode or QR code sent via email or displayed on the hotel app, which automatically fetches reservation details.
- **Pay via Barcode/QR Code**: Customers can scan a QR code to process payments directly from their mobile wallet (e.g., Apple Pay, Google Pay) or bank card.
- **Payment Processing**: The system will automatically calculate the final bill (room charges, services) and prompt the customer for payment.
- **Payment History**: Customers can view a history of all payments made during their stay, including any additional services or room upgrades.
- **Rate Services**: Customers can rate the quality of various services (room, cleanliness, service, food, etc.) using a star-based system or a similar rating scale.
- **Leave Comments**: Customers can leave detailed feedback or suggestions, including both positive and negative experiences, for the hotel's review.
- **Currency Selection**: Customers should be able to view room rates and make payments in their preferred currency.
- **Secure Account Management**: Customers should be able to securely log in, change their passwords, and manage privacy settings to protect personal data.
- **Track Rewards**: Customers should be able to view and redeem their earned loyalty points or rewards (e.g., discounts, free upgrades, free nights).
- **View Reviews**: Customers should be able to read reviews from previous guests to help with their decision-making process during the booking process.
- **Messaging**: The system should allow customers to communicate directly with the hotel (e.g., ask questions, make requests) via chat or email.
- **Request Additional Services**: Customers can request services like extra towels, early check-in, late check-out, transportation, etc.
- **Invoice Generation**: Customers should receive an electronic or paper invoice at check-out, detailing room charges, taxes, and any additional services.
- **Self Check-in/Check-out**: Customers should be able to check-in and check-out online or through kiosks for convenience, especially if they prefer minimal interaction at the front desk.

4.3.3 Receptionist

- **Manage Inventory:** Receptionist can keep track of supplies such as room amenities, office materials, and other consumables.
- **Cancel a Booking:** Receptionist can search for a booking and delete or cancel it if necessary (e.g., customer cancellation or changes).
- **Check Room Availability:** Receptionist can quickly check room availability in real-time, including status (e.g., booked, available, out of service).
- **Customer Registration:** The receptionist can input new customer details into the system (name, contact information, preferences, etc.).
- **Store Customer Profiles:** Ensure that customer profiles are automatically created and stored securely in the system for future bookings or inquiries.
- **Payroll Management:** Receptionist can access and update payroll information for staff members, including salary, bonuses, and deductions.
- **User Authentication:** Receptionist must log into the system using secure authentication methods (username and password, 2FA, etc.).
- **Respond to Emails:** Reply to customer emails, especially regarding bookings, inquiries, or any customer service-related issues.
- **Request Housekeeping**: The receptionist should be able to request housekeeping services, such as cleaning, laundry, or extra linens for a guest's room.
- **Maintenance Requests**: If guests report issues with their rooms (e.g., plumbing, air conditioning), the receptionist should log and escalate maintenance requests.
- **No-Show Management**: The receptionist should be able to mark reservations as "no-show" and update the room availability accordingly.
- **Access Guest Information**: The receptionist should have access to guest profiles (e.g., name, contact information, preferences) to provide a personalized experience.
- **Special Room Requests**: Receptionists should be able to handle specific room requests, such as the need for extra beds, crib, or non-smoking rooms.
- **Notifications**: The receptionist should ensure that guests are informed of any important hotel events, updates, or changes (e.g., Wi-Fi issues, breakfast hours).
- **Room Status Update**: Once checked out, the receptionist should update the room status.

4.3.4 Payment Handler

- **Generate Invoices:** Automatically generate invoices for customers once a payment is processed, including the breakdown of charges (e.g., room cost, additional services, taxes)
- **Payment Processing:** The payment handler can process payments through various methods (e.g., credit/debit cards, bank transfers, online payment gateways like PayPal or Stripe).
- **Refund Processing:** The payment handler can initiate refunds for canceled bookings or returned products/services, updating the system accordingly.
- **Customizable Billing:** Allow the payment handler to adjust bills based on customer needs (e.g., adding discounts, adjusting quantities, applying promotions).
- **Automatic Payment Confirmation:** Once the payment is processed, the system automatically sends a payment confirmation to the customer via email or SMS, including the payment amount, payment method, date, and confirmation number.
- **Automatic Tax Calculation:** The system should automatically calculate applicable taxes (e.g., sales tax, VAT) based on the customer's location, product/service type, and local tax regulations.
- **Apply Discounts:** The payment handler can apply various discounts or promotional codes (e.g., seasonal discounts, loyalty discounts) to customer bills.

- **Track Total Revenue:** The system should allow the payment handler to calculate and track the total revenue generated over a given period (e.g., daily, weekly, monthly).
- **Secure Card Processing:** The system should securely handle card details via payment gateways, ensuring PCI DSS compliance and data encryption.

4.4 Non-Functional Requirement

1. Performance

- **Response Time**: The system should have a fast response time (e.g., under 3 seconds for user actions like booking or check-in).
- **Scalability**: The system should handle increasing numbers of users, bookings, and data efficiently without degradation in performance.

2. Availability

- **High Availability**: The system should be available 24/7 with minimal downtime for maintenance, ensuring continuous access for customers and staff.
- **Disaster Recovery**: Ensure data backup and recovery mechanisms to restore the system in case of failure or data loss.

3. Usability

- **User-Friendly Interface**: The system should have an intuitive and easy-to-use interface for both customers and hotel staff.
- **Mobile Compatibility**: The system should be compatible with mobile devices, providing access via apps or mobile-friendly websites.

4. Localization

- **Currency Support**: It should handle multiple currencies for international payments and billing.

5 Design and Implementation Constraints

5.1 Use case diagram

Figure 5.1- 1 Use case diagram for Hotel management system

5.2 Activity diagram and Swimlane diagram

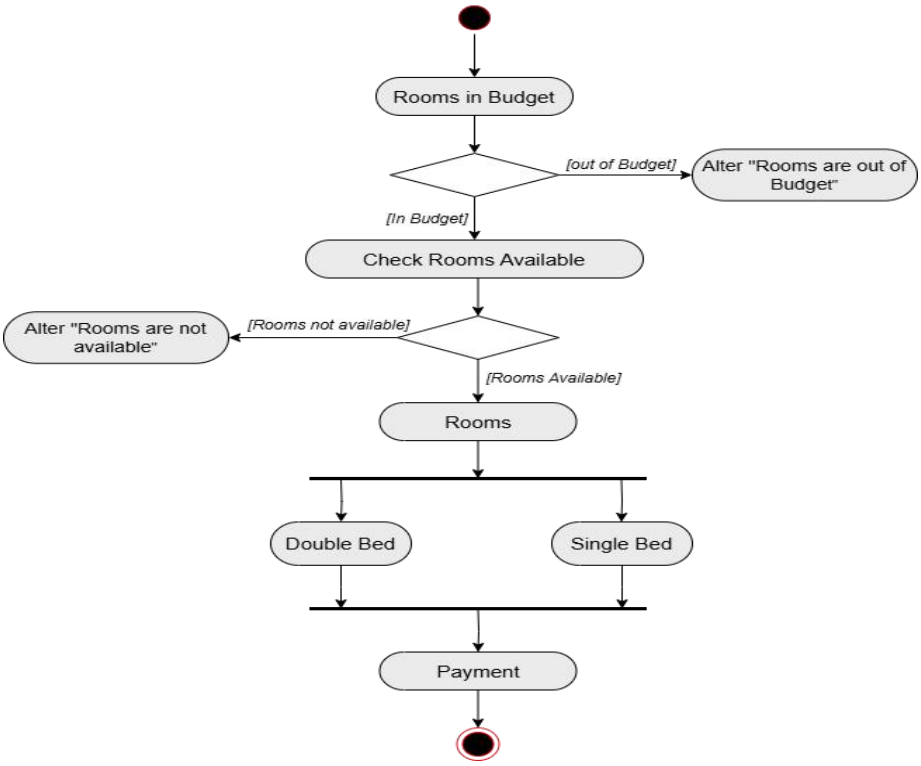

Figure 5.2- 1 Activity diagram for Room Booking

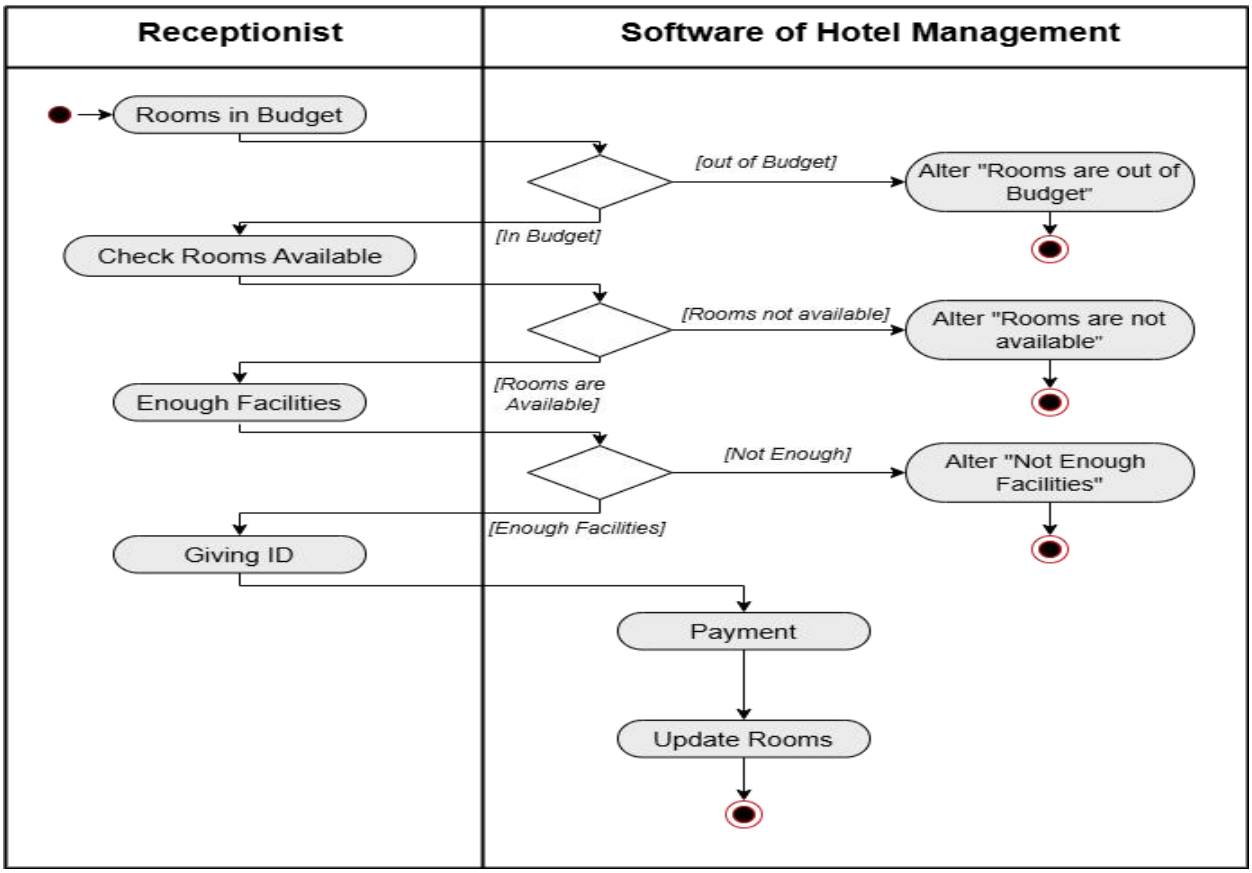

Figure 5.2- 2 Swimlane diagram for Room Booking

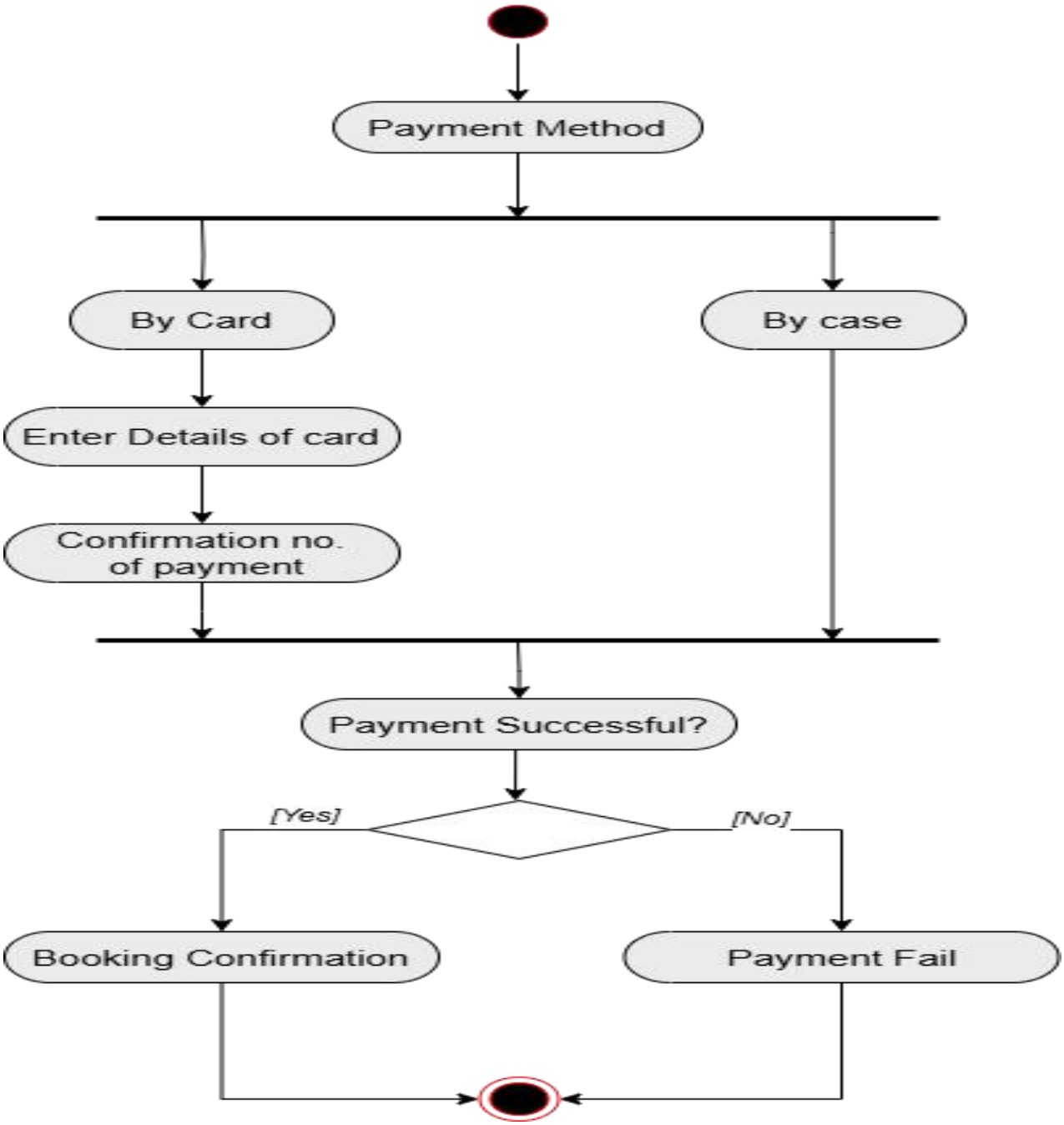

Figure 5.2- 3 Activity diagram for Payment

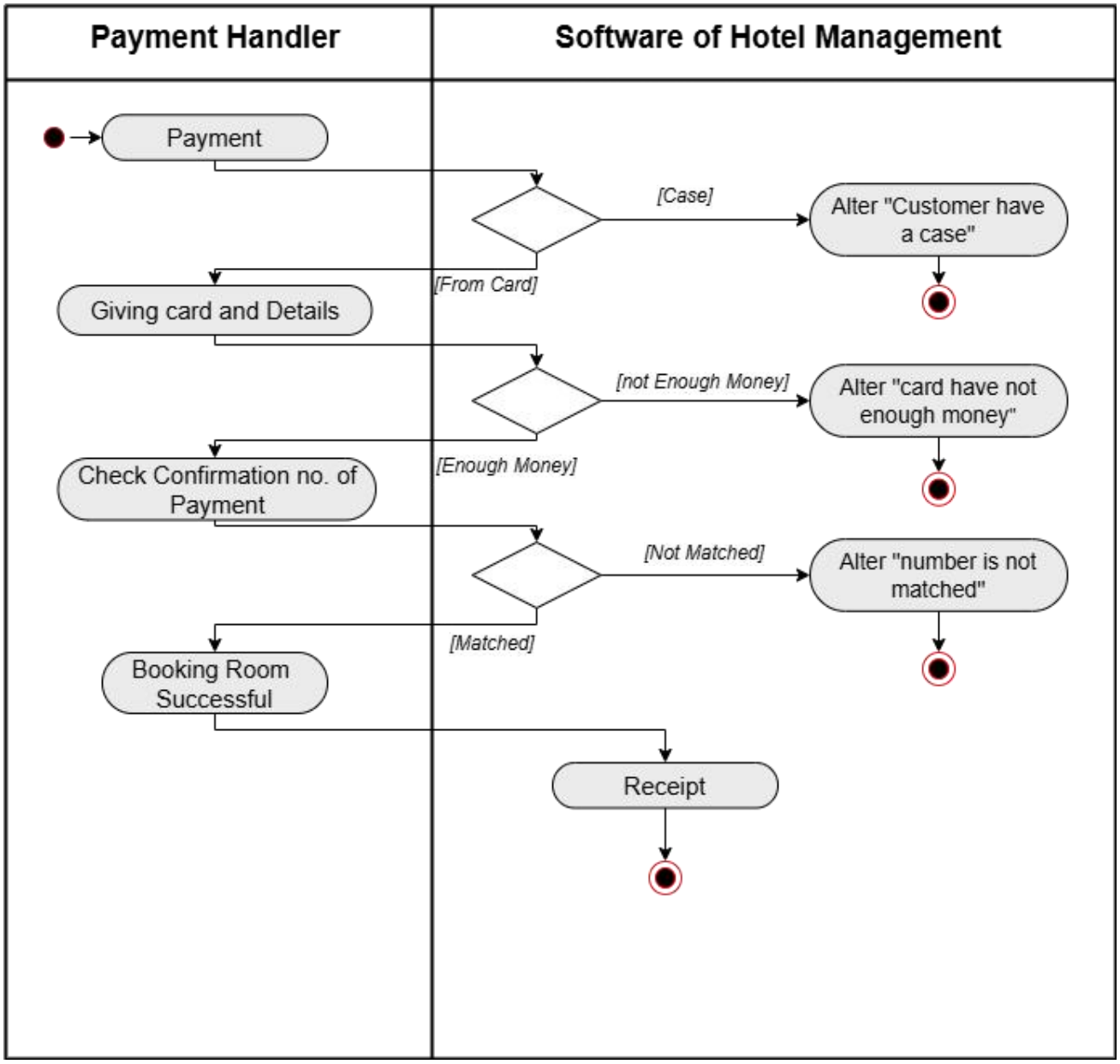

Figure 5.2- 4 Swimlane diagram for Payment

5.3 Sequence diagram

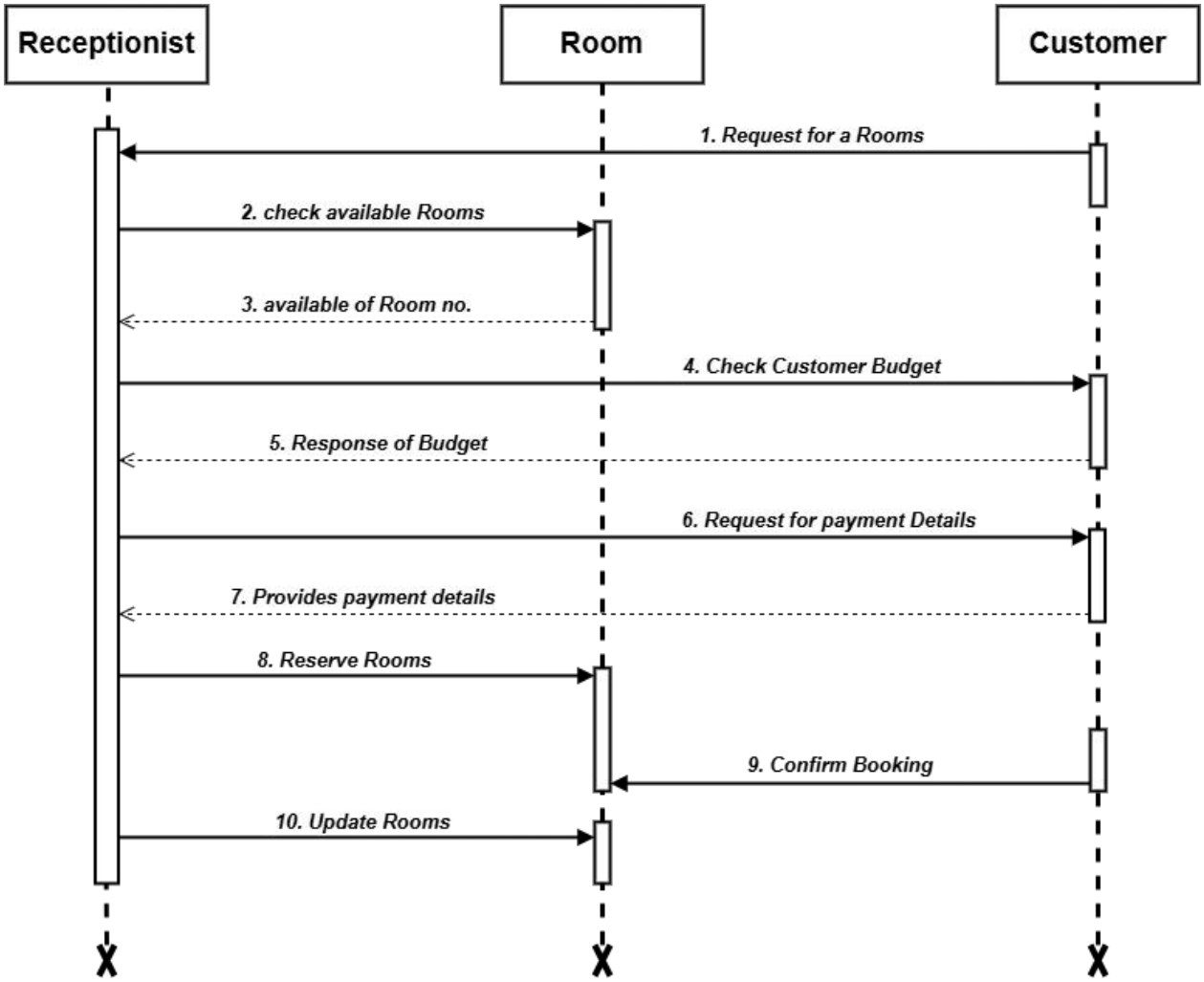

Figure 5.3- 1 Sequence diagram for Room Booking

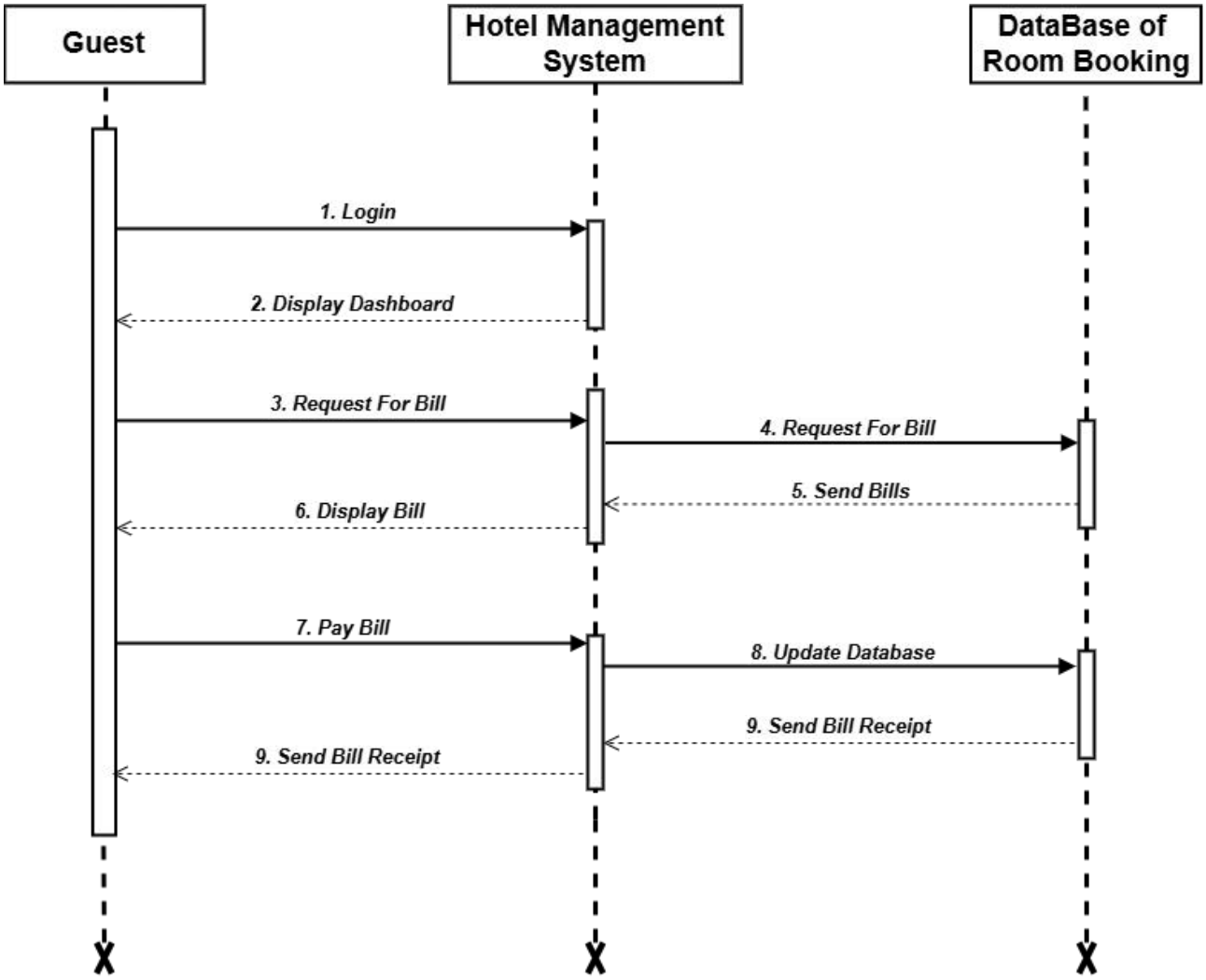

Figure 5.3- 2 Sequence diagram for Payment

5.4 State diagram

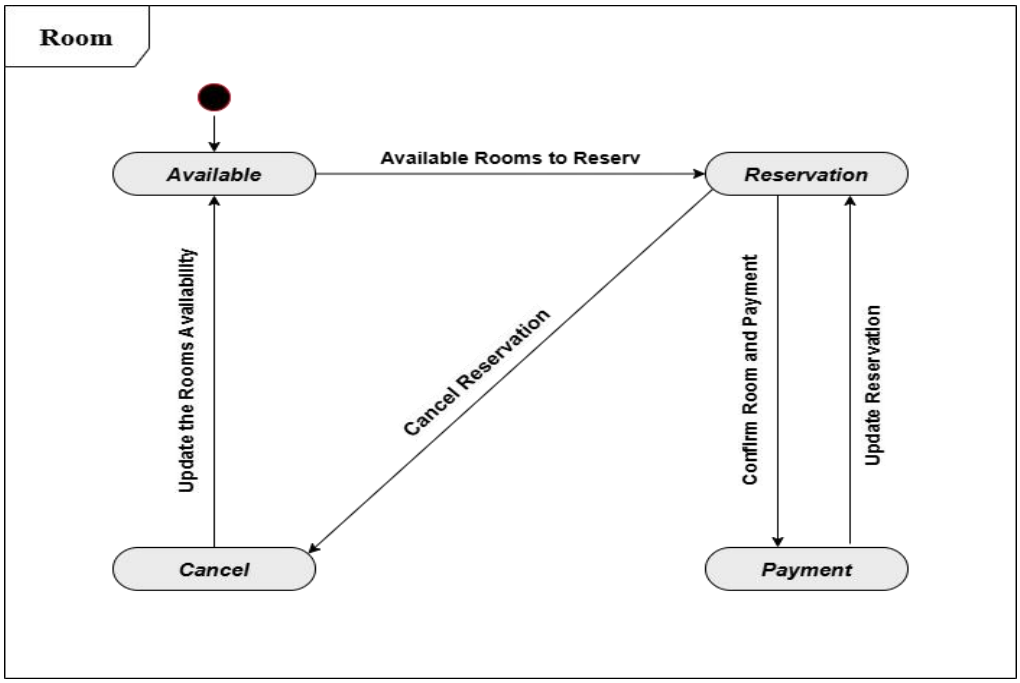

Figure 5.4- 1 State diagram of Room

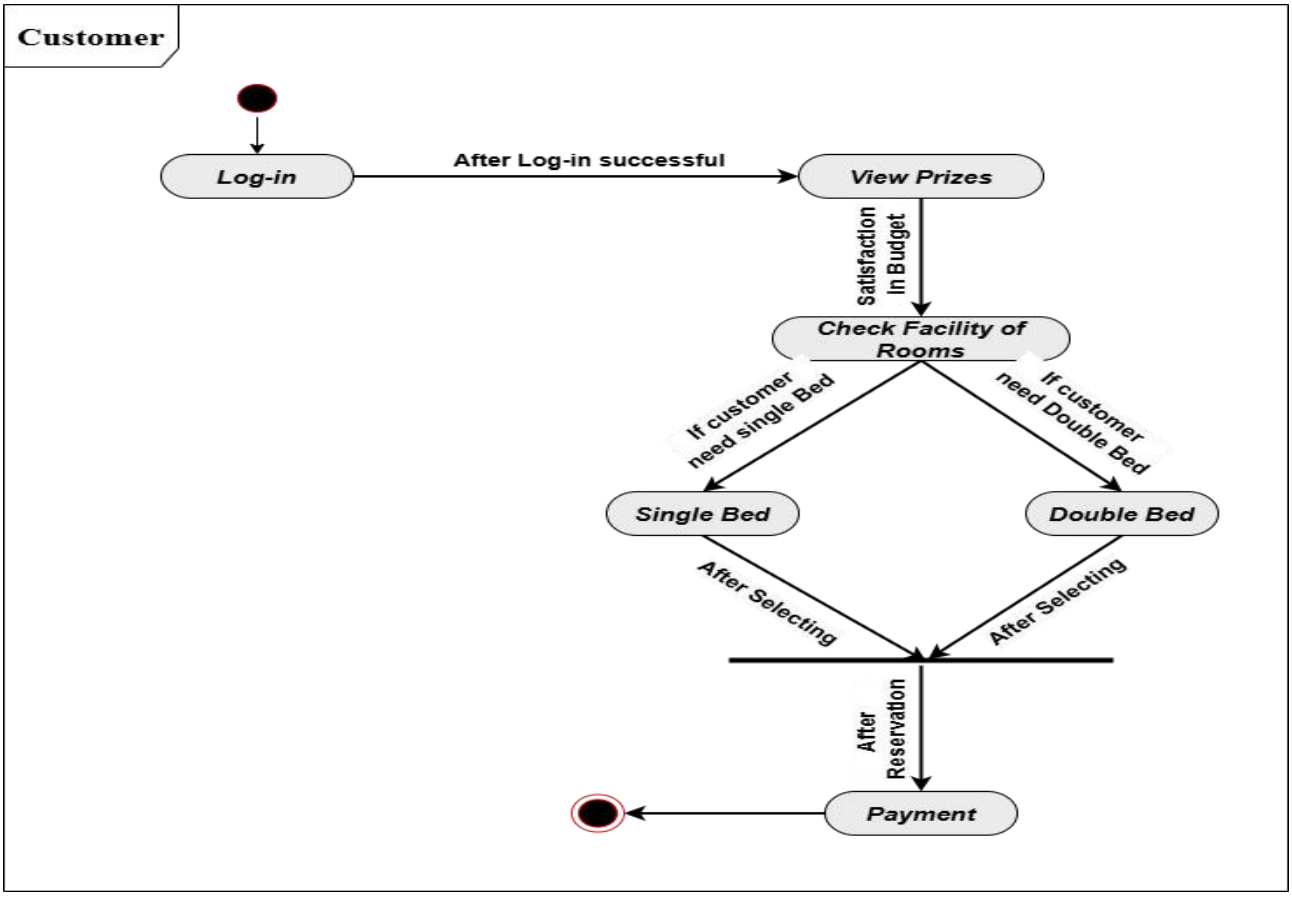

Figure 5.4- 2 State diagram for Customer

5.5 Class diagram

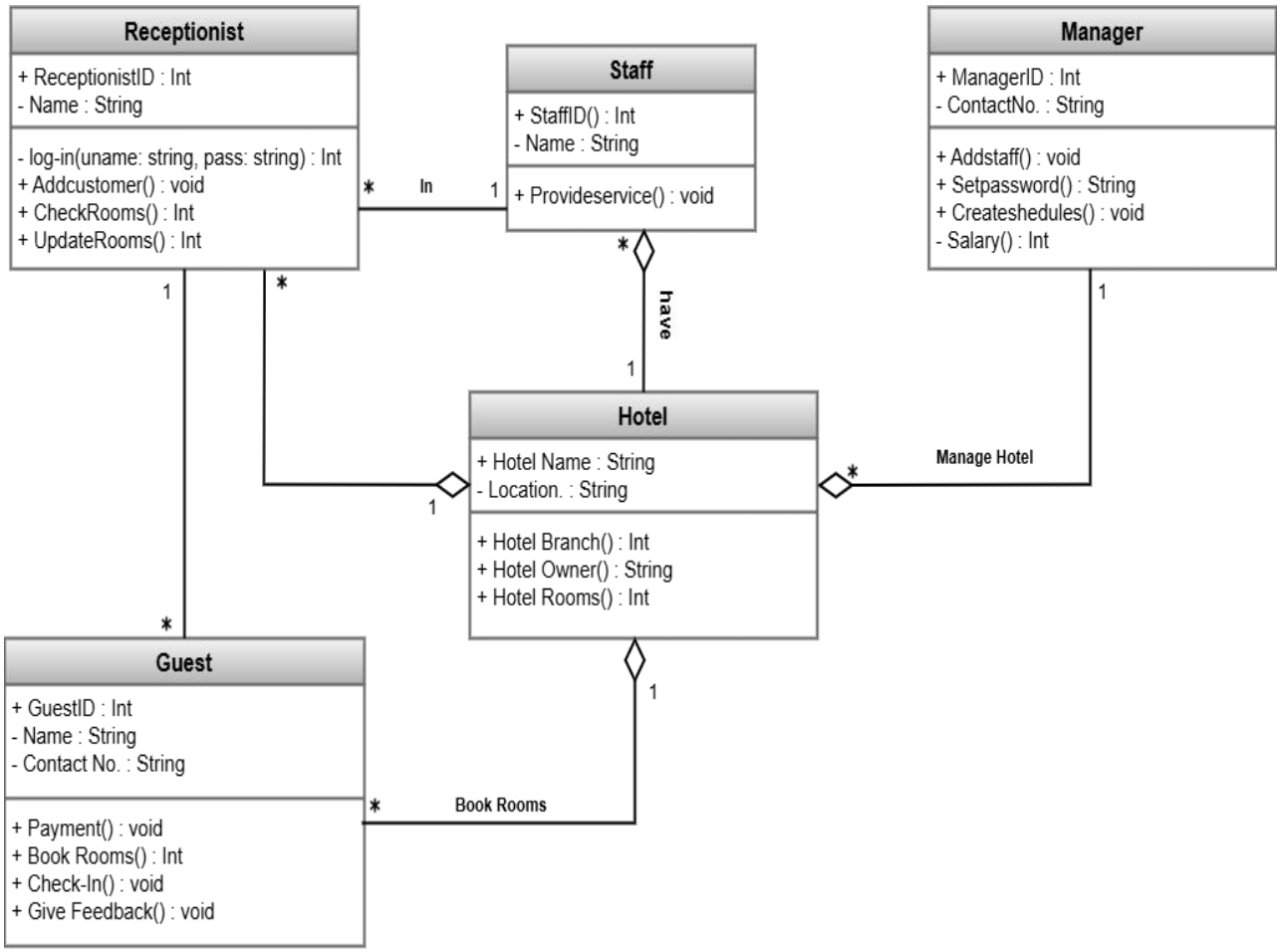

Figure 5.5- 1 Class diagram for Hotel management system

5.6 Data flow diagram

5.6.1 Context diagram (level-0)

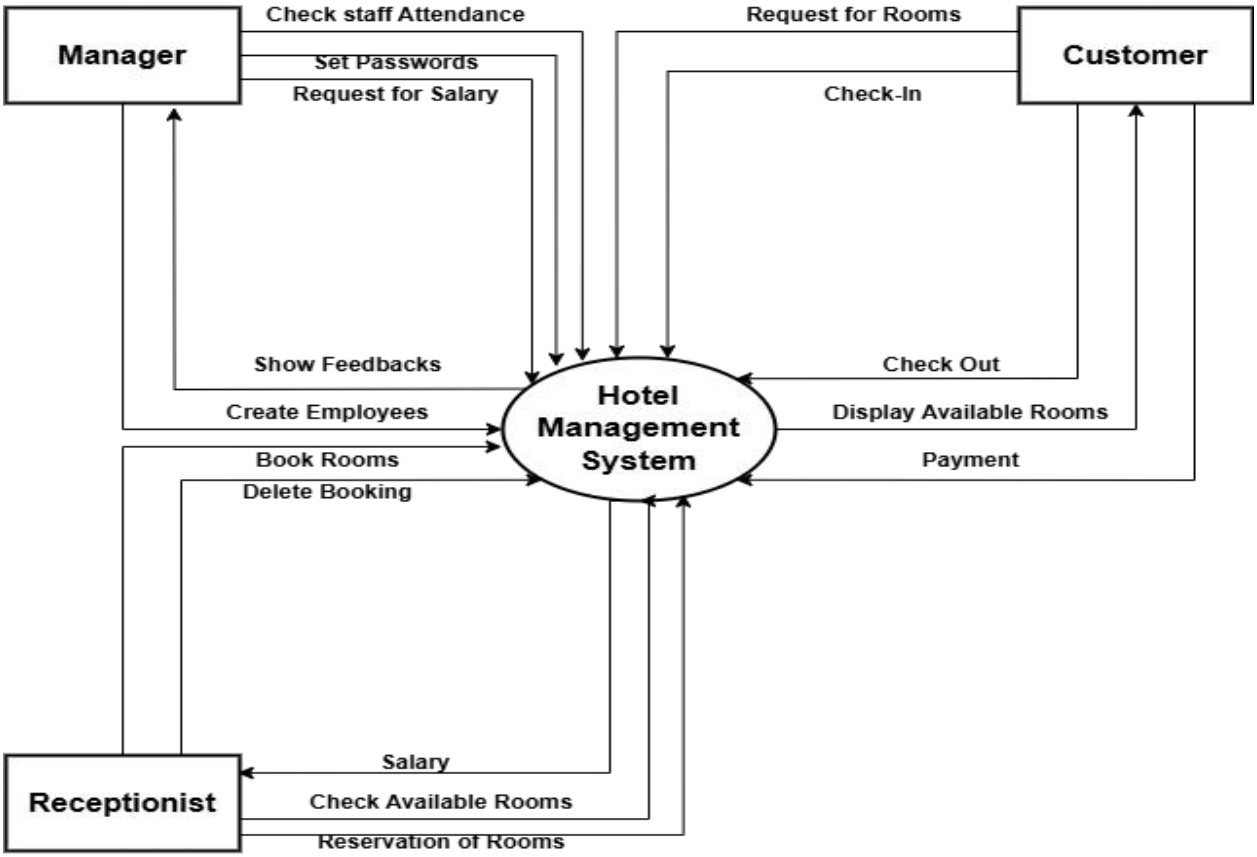

Figure 5.6- 1 Context diagram for Hotel management system

5.6.2 DFD Level-1

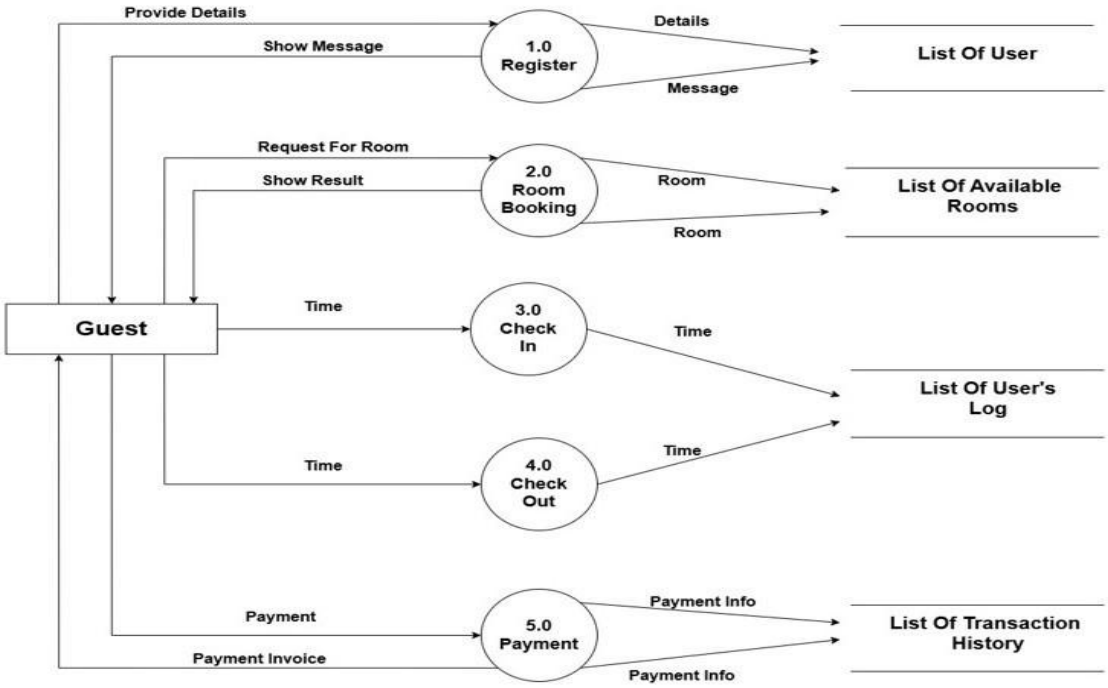

Figure 5.6- 2 DFD level-1 for Hotel management system

6 External interface requirement (Screens)

6.1 Screen-1: Create Employees

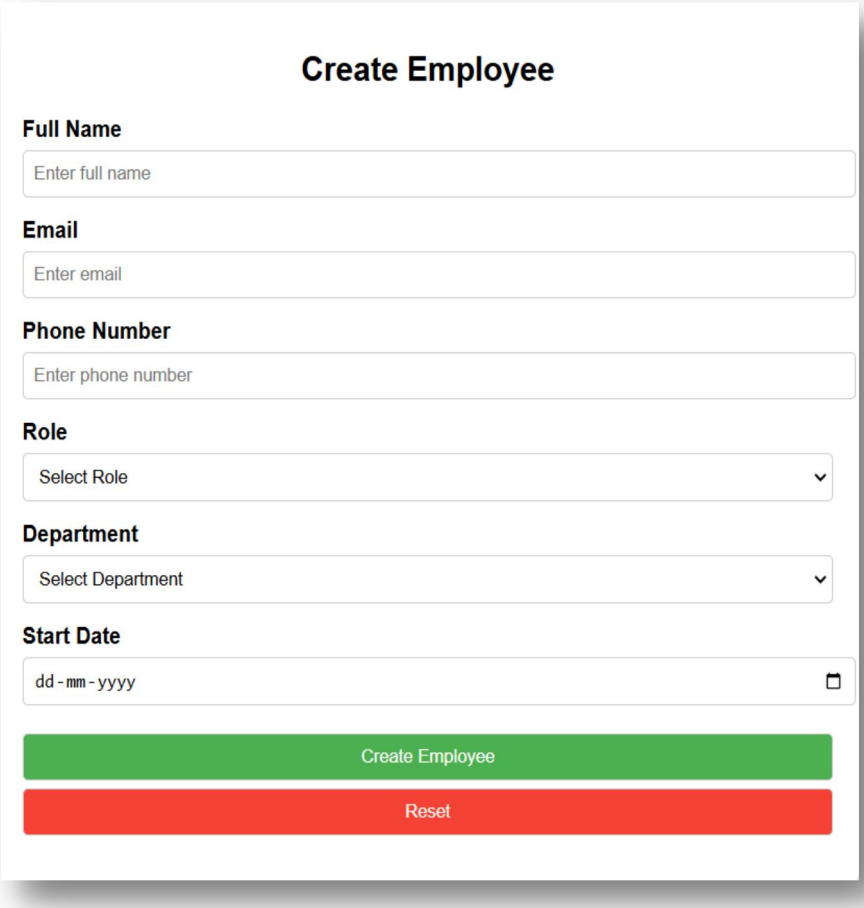

Figure 6.1- 1 Screen-1: Create Employees

Table 6.1- 1 Screen element of Create Employees

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Full Name	Textbox	M	1	Enter the employee's full name.
2	Email	Textbox	M	1	Enter a email address.
3	Phone Number	Textbox	M	1	Enter a phone number.
4	Role	Dropdown	M	1	Select the role of the employee.
5	Department	Dropdown	M	1	Choose the department.
6	Start Date	Date	M	1	Select the employee's start date.
7	Submit Button	Button	M	1	Create employee record with entered details.
8	Reset Button	Button	M	1	Reset all fields in the form.

6.2 Screen-2: Set Password

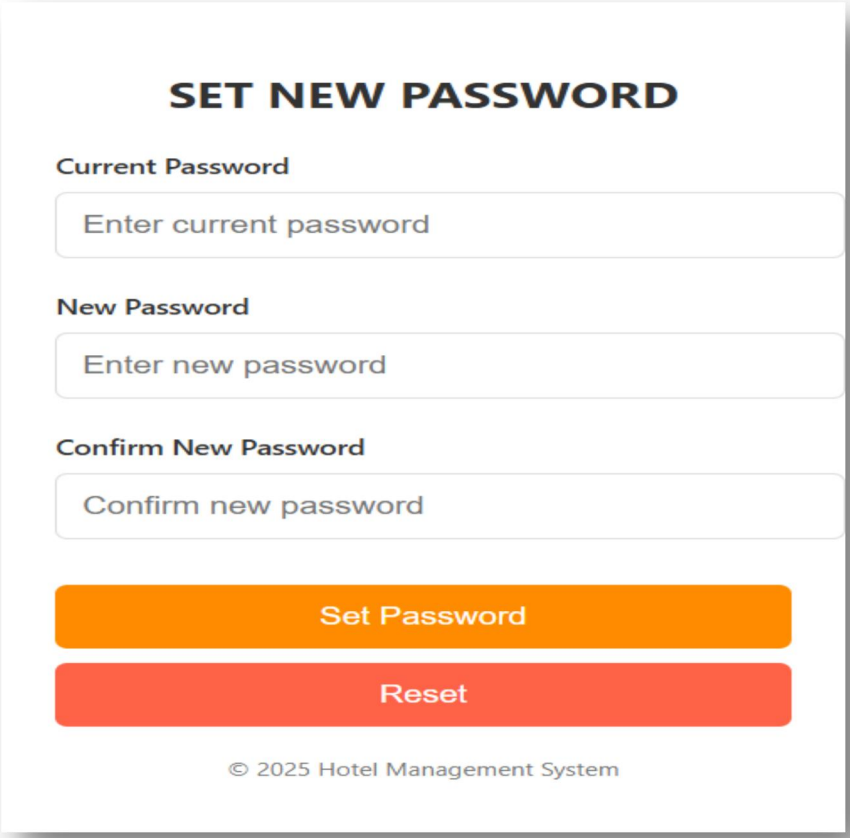

Figure 6.1- 2 Screen-2: Set password

Table 6.1- 2 Screen element of Set Password

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Current Password	Password	M	1	Enter the current password.
2	New Password	Password	M	1	Enter the new password.
3	Confirm New Password	Password	M	1	Confirm the new password.
4	Set Password	Button	----	----	Button for set the new password.
5	Reset	Button	----	----	Button for reset the fields.

6.3 Screen-3: Customer using Barcode Scanner

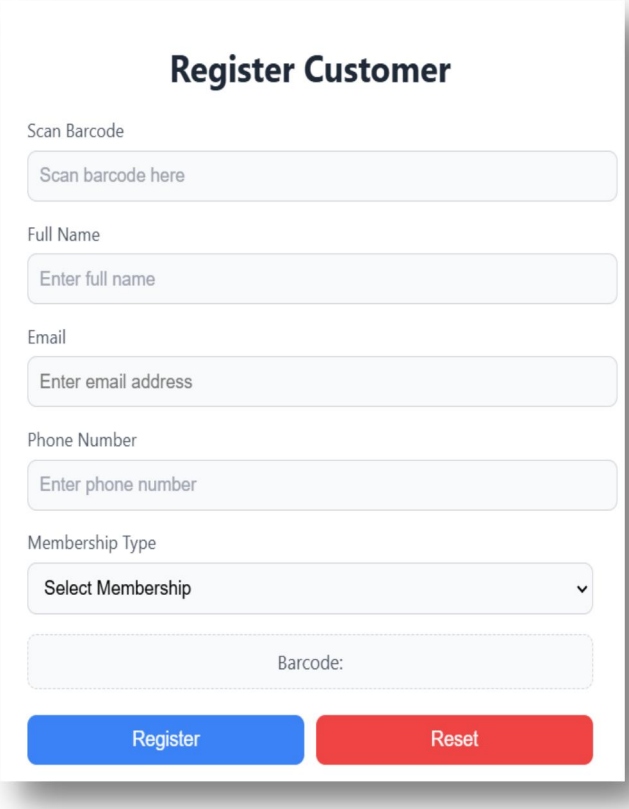

Figure 6.1- 3 Screen-3: Add Customer

Table 6.1- 3 Screen element of Add Customer

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Scan Barcode	Textbox/Scanner Input	M	1	Scan or Enter the barcode.
2	Full Name	Textbox	M	1	Enter the full name of the customer.
3	Email	Textbox	M	1	Field to enter the email address of the customer.
4	Phone Number	Textbox	M	1	Field to enter the phone number of the customer.
5	Membership Type	Dropdown	M	1	Select the membership type.
6	Barcode	Label	----	----	Displays the barcode information.
7	Register	Button	----	----	Button for register the customer.
8	Reset	Button	----	----	Button for reset the form fields.

6.4 Screen-4: Payment Processing for Customer:

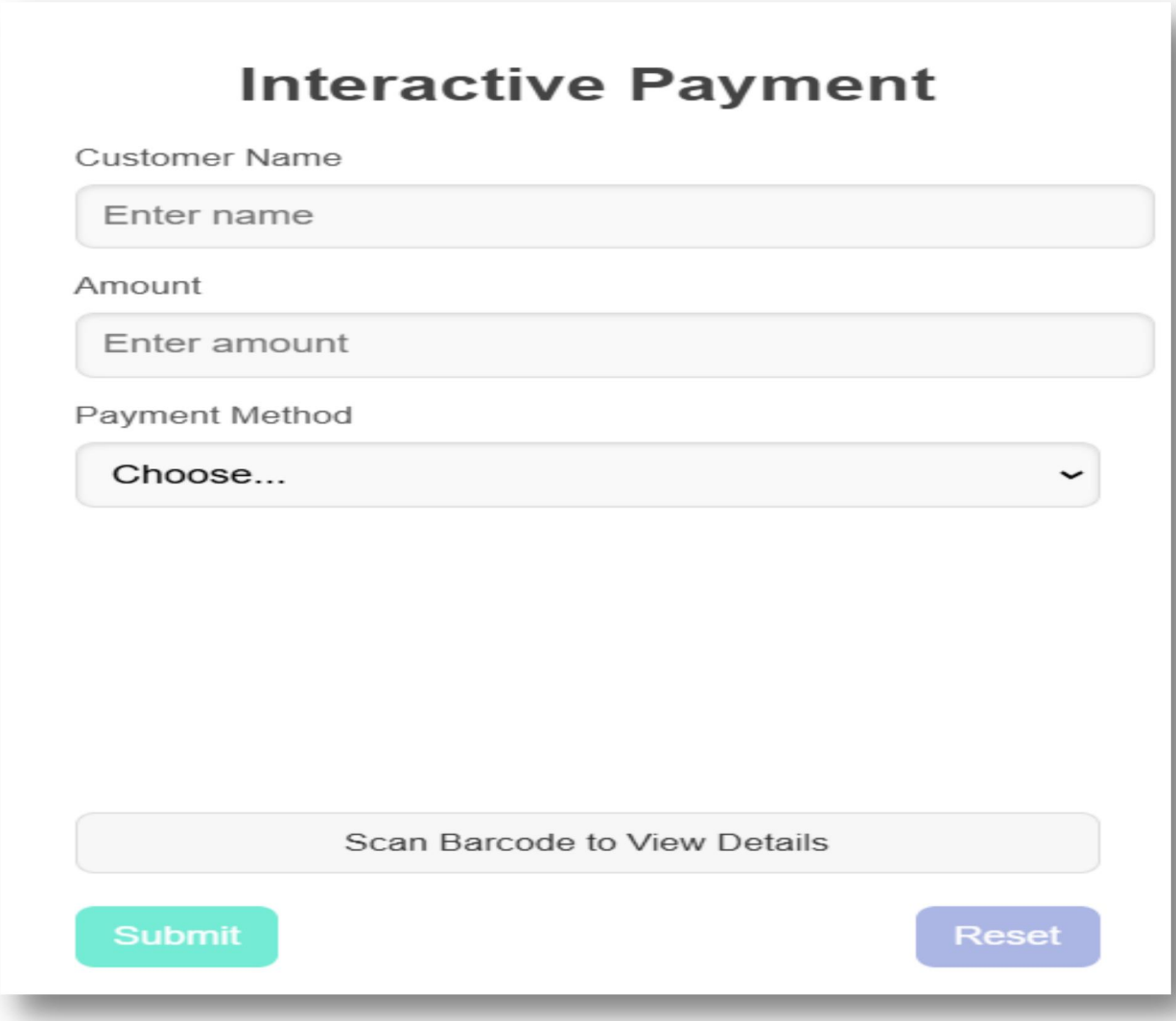

Figure 6.1- 4 Screen-4: Payment Processing

Table 6.1- 4 Screen element of Payment

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Customer Name	Textbox	M	1	Enter the customer's name.
2	Amount	Textbox	M	1	Enter the payment amount.
3	Payment Method	Dropdown	M	1	Select the payment method.
4	Scan Barcode to View Details	Scanner Input	M	1	Scan barcode for viewing details.
5	Submit	Button	----	----	Button for submit the payment details.

6	Reset	Button	----	----	Button for reset the form fields.

6.5 Screen-5: Cancel a Booking:

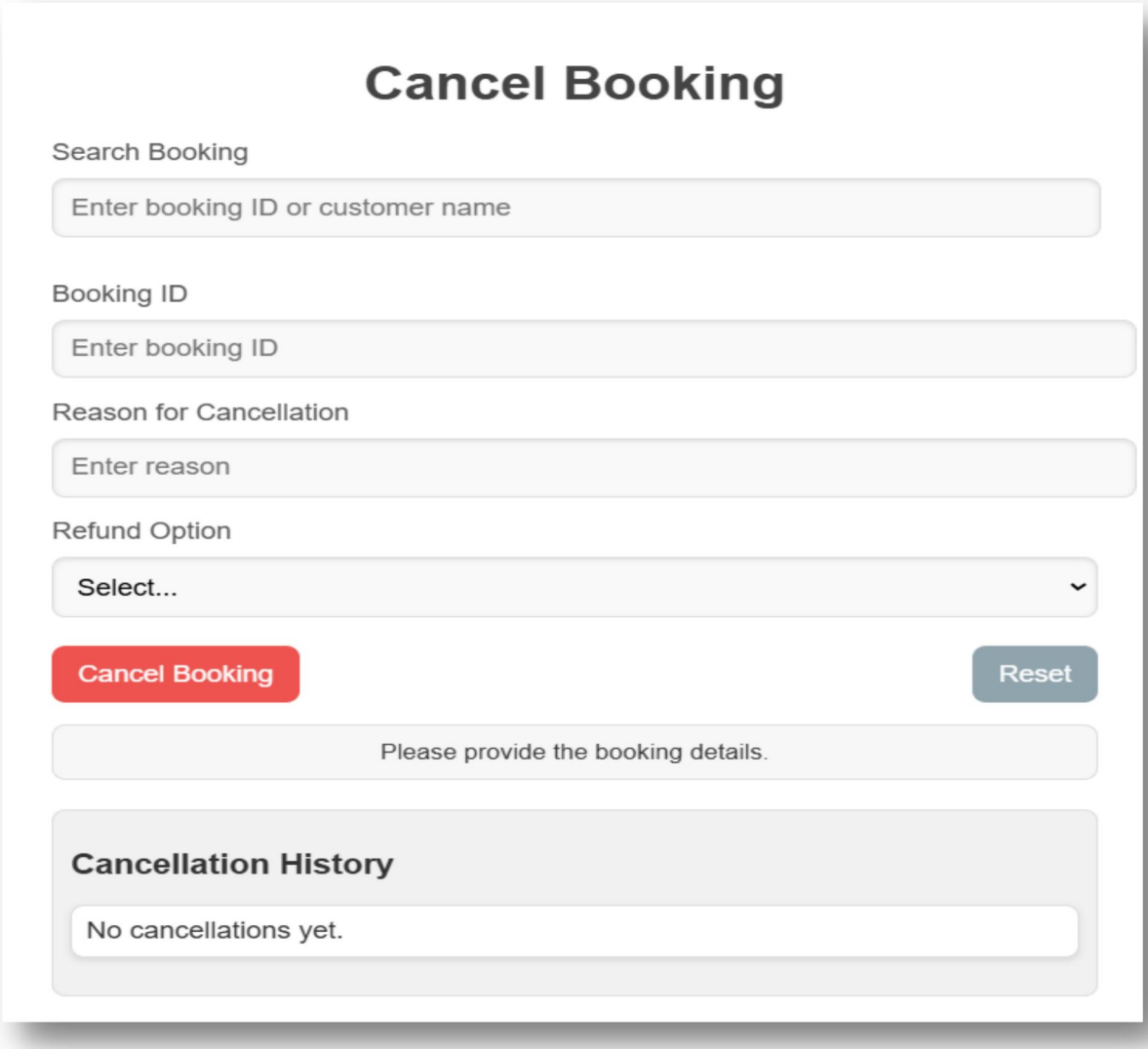

Figure 6.1- 5 Screen-5: Cancel Booking

Table 6.1- 5 Screen element Cancel Booking

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Search Booking	Textbox	M	1	Enter booking ID or customer name for searching.
2	Booking ID	Textbox	M	1	Enter the booking ID.
3	Reason for Cancellation	Textbox	M	1	Enter the reason for cancellation.
4	Refund Option	Dropdown Button	M	1	Select the refund method.

5	Cancel Booking	Button	----	----	Button for Cancel a customer Booking
5	Reset	Button	----	----	Button for reset the form fields.
6	Please provide the booking details.	Label	----	----	User to provide booking details.
7	Cancellation History	Label	----	----	Section heading for cancellation history.
8	No cancellations yet.	Label	----	----	Show available cancellation history.

7 Database design

7.1 List of Tables

- **Room**
- **Booking**
- **Customer**
- **Payment**
- **Hotel**

Table 7.1- 1 Table: Room

Column	Data Type	Null	Keys & Constraints	Default Value & Description
Room_Number	Int	NN		Number of the room.
Room_Price	Int	NN		Price of the room.
Room_Type	varchar(100)	NN		Type of the room like Double or Single bed.
Rating	Int	NN		Customer rating for the room
RoomID	Int	NN	PK	Unique identifier for each room record.

Table 7.1- 2 Table: Booking

Column	Data Type	Null	Keys & Constraints	Description
Booking_ID	Int	NN	PK	Unique identifier for each booking record.
User_ID	Int	NN	FK	ID of the User.
Room_ID	Int	NN	FK	ID of Room
Check_in_date	Date&Time	AN		Date and time of check-in for the booking.
Check_out_date	Date&Time	AN		Date and time of check-out for the booking.
Booking_status	Varchar(100)	NN		Current status of the booking

Table 7.1- 3 Table: Customer

Column	Data Type	Null	Keys & Constrains	Default Value & Description
CustomerID	int	NN	PK (Auto Increment)	Unique identifier for each customer.
CustomerName	varchar(100)	AN		Full name of the customer.
Phone_number	Varchar(10)	AN		Contact number of the customer.
Password	Varchar(10)	NN		Set a password for customer account access.
Email	varchar(100)	AN		Email address of the customer.

Table 7.1- 4 Table: Payment

Column	Data Type	Null	Keys & Constraints	Default Value & Description
PaymentID	int	NN	PK (Auto Increment)	Unique identifier for each payment record.
Amount	Decimal(10,2)	NN		The amount of the payment.
Payment_Date	DateTime	NN		Date and time of the Payment.
Booking_ID	Int	NN	FK	Foreign key for the booking with the payment.
Payment_method	Varchar(100)	AN		Method used for the payment (e.g., Credit Card, PayPal).
Card_number	Varchar(16)	NN		Card number used for the payment.

Table 7.1- 5 Table: Hotel

Column	Data Type	Null	Keys & Constraints	Default Value & Description
HotelID	int	NN	PK (Auto Increment)	Unique identifier for each hotel record.
Hotel_Name	Varchar(100)	NN		Name of the hotel.
City	Varchar(100)	NN		City of the Hotel.
Star_Rating	Int	NN		Star rating of the hotel.
Address	Varchar(100)	AN		Address of the hotel.

8 Stories and Scenario

8.1 Story-1: Book a Room in Hotel Catalogue

*Story # **S1***	:	**As a** Guest, **I want to** rent a room in Hotel catalogue **So that** I can rest and stay in comfort.
Priority	:	High
Estimate	:	XL
Reason	:	The ability to rent a room directly from the hotel catalogue ensures a seamless booking process. It allows for immediate access to room options, ensuring that guests can easily find and book a room that fits their preferences.

8.1.1 Scenario# S1.1

*Scenario# **S1.1***	:	Booking a Room with Valid Information
Prerequisite	:	Guest is logged in to the Hotel management system.
Acceptance Criteria	:	**Given:** The Guest is navigated to the hotel room booking page, where available rooms are displayed with details such as room type, pricing, and amenities. **When:** The Guest selects the "Book a room" option And The Guest enters valid personal details The Guest clicks the "Confirm Booking" button and procced with the reservation. **Then** the system successfully book the room to the Hotel and the guest receives a booking confirmation message with the room details, booking reference number, and check-in instructions.

8.1.2 Scenario# S1.2

*Scenario# **S1.2***	:	Booking a Room with Valid Information
Prerequisite	:	Guest is logged in to the Hotel management system.
Acceptance Criteria	:	**Given:** The Guest is navigated to the hotel room booking page, where available rooms are displayed with details such as room type, pricing, and amenities. **When:** The Guest selects the "Book a room" option And The Guest enters valid personal details The Guest clicks the "Confirm Booking" button and procced with the reservation. **Then** The system detects the invalid information and prompts the guest to correct the errors.

8.2 Story-2: Available rooms

*Story # **S2***	:	**As a** Guest, **I want to** view information about available rooms in the hotel catalogue, **So that** I can make an informed decision about which room to book based on my preferences.
Priority	:	High
Estimate	:	M
Reason	:	Providing guests with detailed room information, room layout, pricing is crucial for helping them for select a Room. This will enhance their booking experience by ensuring they select the room that best suits their needs, ultimately improving guest satisfaction.

8.2.1 Scenario# S2.1

*Scenario# **S2.1***	:	Viewing Details of Available rooms
Prerequisite	:	Guest is logged in to the Hotel management system.
Acceptance Criteria	:	**Given:** The Guest is on the room catalogue page, which displays available rooms with details such as room name, price. **When:** The guest selects a room they are interested in from the list of available rooms. The guest clicks the "View Details" button next to the selected room. **Then** system displays the detailed information for the selected room The guest can either proceed with booking the room or return to the room catalogue to view other options.

8.2.2 Scenario# S2.2

*Scenario# **S2.2***	:	Viewing Details of Available rooms
Prerequisite	:	Guest is logged in to the Hotel management system.
Acceptance Criteria	:	**Given:** The Guest is on the room catalogue page, which displays available rooms with details such as room name, price. **When:** The guest selects a room they are interested in from the list of available rooms. The guest clicks the "View Details" button next to the selected room. **Then** system send a message that “the selected room is not available for the selected dates.”

8.3 Story-3: Make Payment

*Story # **S3***	:	**As** Guest, **I want** to make a payment for my room booking, **So that** I can confirm my reservation.
Priority	:	High
Estimate	:	XL
Reason	:	A secure payment process is for to complete a booking. By providing multiple payment methods that guests can easily make a payment.

8.3.1 Scenario# S3.1

*Scenario# **S3.1***	:	Making a Payment for Room Booking
Prerequisite	:	Guest must select a room and is on the payment page of the hotel booking system.
Acceptance Criteria	:	**Given:** The Guest has selected a room and entered all booking details. **When:** the guest enters payment details and clicks the "Pay Now" button, **Then** the system should validate the payment details, process the transaction securely, And the system should display a success message with a confirmation number.

8.3.2 Scenario# S3.2

*Scenario# **S3.2***	:	Confirming the Booking
Prerequisite	:	The guest must have entered their payment information and is ready to confirm the booking.
Acceptance Criteria	:	**Given:** The guest has entered their payment information. **When:** They click the "Confirm Booking" button, **Then:** the system should display a confirmation message with the booking details. And the guest should receive a confirmation email with the same details.

9 Test cases

Project Name:	Hotel Management System	Test Designed by:	Pujan Patel
Module Name:	Room Reservation	Test Designed date:	18-02-2025
Release Version:	1.0	Test Executed by:	R. B. Gondaliya
		Test Execution date:	15-01-2023

Pre-condition: Web application should be accessible				
Test Case ID	**Test Title**	**Test Type**	**Description**	**Test Case ID**
TC_001	Room Reservation with Valid Details	Functional	Reserve a room in the hotel with valid details such as check-in/check-out dates, room type, and details.	TC_001
TC_002	Room Reservation with Invalid Dates	Functional	Try to reserve a room with invalid check-in/check-out dates	TC_002

Test Case Title	Room Reservation with Valid Details
Test Type	Functional
Test Priority	High
Pre-condition	Web application should be accessible

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Access Hotel reservation website URL	The site should load properly without errors	Site launched successfully	Pass		https://accounts.google.com/ServiceLogin	
2	Enter guest details (name, email, etc)	Guest details fields should be editable and accept valid inputs.	Guest details accepted	Pass		Name: Pujan Pipaliya, Email: pujan@email.com	
3	Choose room type (single, double etc.)	Room type selection should be available and show available rooms for the dates selected.	Room Type selected correctly	pass		Room Type: Double Bed	
4	Verify confirmation details	The confirmation page should display correct reservation details	Details correct	Pass		Room type: Double Room, Dates: 2025-03-01 to 2025-03-05	

Test Case Title	Room Reservation with Invalid Dates
Test Type	Functional
Test Priority	Medium
Pre-condition	Web application should be accessible

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Select valid check-in and check-out dates	The system should accept valid dates for check-in and check-out fields.	Dates Selected correctly	Pass		Check-in: 2025-03-01, Check-out: 2025-03-05	
2	Enter invalid guest details	The system should flag invalid or missing details and prompt the user to correct them.	Error message displayed	Pass		Name: pujan patel, Email: not valid email	
3	Enter an invalid number of guests (e.g., non-numeric or too many)	The system should display an error message indicating the number of guests is invalid.	Error message displayed	Pass		Guest: -1	
4	Verify that the reservation does not proceed	The system should prevent the user from completing the reservation until all details are corrected.	Reservation not completed	Pass			

Project Name:	**Hotel Management System**	**Test Designed by:**	**Pujan Patel**
Module Name:	**Payment Process**	**Test Designed date:**	18-02-2025
Release Version:	**1.0**	**Test Executed by:**	**R. B. Gondaliya**
		Test Execution date:	15-01-2023

Pre-condition: Web application should be accessible				
Test Case ID	**Test Title**	**Test Type**	**Description**	**Test Case ID**
TC_001	Payment Process for Room Reservation	Functional	Process the payment for a room reservation, including selecting a payment method and confirming payment.	TC_001
TC_002	Make Payment with Invalid Card Details	Functional	Attempt to make a payment using invalid card details and verify that the system rejects the payment with an error message.	TC_002

Test Case Title	Payment Process for Room Reservation
Test Type	Functional
Test Priority	High
Pre-condition	Web application should be accessible

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Proceed to the payment section	The user should be directed to a payment section with the booking details displayed.	Payment page displayed	Pass			
2	Choose a payment method (credit card, PayPal, etc.)	The system should show available payment methods	Payment method displayed	Pass		Payment method: Credit Card	
3	Enter payment details	The system should securely accept and process the payment details entered.	Payment details accepted	pass		Card Number: 1234 5678 9876 5432,	

4	Verify reservation details on the confirmation page	The confirmation page should display correct reservation details	Details correct	Pass		Room type: Double Room, Dates: 2025-03-01 to 2025-03-05	

Test Case Title	Payment Attempt with Invalid Card Details
Test Type	Functional
Test Priority	High
Pre-condition	Web application should be accessible

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Proceed to the payment section	The user should be directed to a payment section with the booking details displayed.	Payment page displayed	Pass			
2	Choose a payment method (credit card, PayPal, etc.)	The system should show available payment methods	Payment method displayed	Pass		Payment method: Credit Card	
3	Enter Invalid payment details	The system should reject the payment and display an error message.	Error Message Displayed	pass		Card Number: 0000 0000 0000 0000,	
4	Verify that the reservation is not confirmed	The reservation should remain unconfirmed due to the failed payment.	Reservation not confirmed	Pass			

10 References

- http://www.w3schools.com/html/html_intro.asp
- https://www.w3schools.com/php/default.asp
- https://www.javatpoint.com/uml

demo

GET Get

StartFragment

| Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `lon` | required | Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `appid` | required | Your unique API key (you can always find it on your account page under the "API key" tab) | | `exclude` | optional | By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces).

Available values:

- `current`

- `minutely`

- `hourly`

- `daily`

- `alerts` | | `units` | optional | Units of measurement. |

EndFragment

PUT Put

https://api.openweathermap.org/data/3.0/onecall?lat={lat}&lon={lon}&exclude={part}&appid={01c9a54ab909656b46f50a8b7f8910b0}

StartFragment

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EndFragment

PARAMS

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appid	{01c9a54ab909656b46f50a8b7f8910b0} Your unique API key (you can always find it on your account page under the "API key" tab)

PATCH Patch

https://api.openweathermap.org/data/3.0/onecall?lat={lat}&lon={lon}&exclude={part}&appid={01c9a54ab909656b46f50a8b7f8910b0}

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DELETE Delete

https://api.openweathermap.org/data/3.0/onecall?lat={lat}&lon={lon}&exclude={part}&appid={01c9a54ab909656b46f50a8b7f8910b0}

StartFragment

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PARAMS

lat

{lat}

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{01c9a54ab909656b46f50a8b7f8910b0}

Your unique API key (you can always find it on your account page under the "API key" tab)

HEAD Head

https://api.openweathermap.org/data/3.0/onecall?lat={lat}&lon={lon}&exclude={part}&appid={01c9a54ab909656b46f50a8b7f8910b0}

StartFragment

| Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `lon` | required | Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `appid` | required | Your unique API key (you can always find it on your account page under the "API key" tab) | | `exclude` | optional | By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces).

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EndFragment

PARAMS

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{lat}

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lon

{lon}

Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API

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{part}

By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces). Available values:

appid

{01c9a54ab909656b46f50a8b7f8910b0}

Your unique API key (you can always find it on your account page under the "API key" tab)

POST Post

StartFragment

| Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `lon` | required | Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API | | `appid` | required | Your unique API key (you can always find it on your account page under the "API key" tab) | | `exclude` | optional | By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces).

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